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Nasal Aspirator Capable of Automatically Relieving Pressure and Method for Sucking Nasal Mucus

Abstract

A nasal aspirator capable of automatically relieving pressure comprises casing, air pump component, air bag, constant pressure device and suction component; wherein, the air pump component is communicated with the air bag, and the air bag is communicated with the constant pressure device; when the air pump component works, negative pressure will be generated in the air bag; when the pressure exceeds a threshold, a pressure releasing valve will be opened to keep the air pressure inside the air bag stable; when the pressure is less than the threshold, the pressure releasing valve will be closed; and the suction component is communicated with the air bag; it further comprises a switch component, wherein the suction component defines a path for external air to enter the air bag, and the suction component is also provided with a pressure releasing channel, and the switch component can open or close the channel.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to the technical field of nasal aspirators, in particular to a nasal aspirator capable of automatically relieving pressure and a nasal mucus suction method.

BACKGROUND

[0002] Nasal aspirators are usually used to clean the nostrils of people or any other animals. For example, infants and young children lack the ability to clean their nostrils efficiently, and nasal congestion will last for a long time, affecting breathing, eating, breastfeeding, sleep and mood. The nasal aspirator can help parents to discharge secretions from the nasal cavity, so that infants can breathe smoothly without being affected by nasal congestion. Therefore, parents may buy various types and functions of nose inhalers. A common nasal aspirator is a rubber ball. The ball has a tip that can be inserted into the baby's nostril. Squeezing the ball can air exhaust the air in the ball through the top end. When the ball is released, a low pressure area (negative pressure area) will be formed in the ball, and air (and possibly mucus) will be brought into the ball through the tip. When the ball is inflated again due to elasticity, the pressure will reach a balance (return to the natural state). This process will generate a suction force from the tip of the nose to suck out the secretions in the nostril (assuming that the tip is placed in the nostril when it is released), but it is not easy to suck out the secretions in the nasal cavity of infants, and parents cannot control the strength when using it, which is very likely to hurt the nasal cavity of infants.

[0003] For example, the patent document U.S. Pat. No. 9,993,584 “Nasal Aspirator” discloses a suction device. A described nasal aspirator system can have accurate, fine-tuned control of suction through the nasal aspirator system. The nasal aspirator system can include a tip inserted into a baby's nostril, with another end inserted into a parent's mouth. The parent can have accurate control of an amount of suction by regulating their own lungs and depth and speed of breath. The described nasal aspirator system is therefore much safer than previous aspirator systems since a parent can precisely control the suction and stop instantaneously (as required). However, this kind of nasal aspirator is more traditional, needs manual control, and cannot control the efficiency, and the user experience is not good enough.

[0004] Another kind of nasal aspirator usually includes a suction head and a bulb body part, the bulb body part is provided with an air pump, the suction head is provided with a cavity, and the cavity wall opening of the cavity forms a suction port. When the dirt is sucked into the cavity, it is easy to block the suction port, and then the dirt is sucked into the main machine such as the air pump, which is easy to cause damage to the main machine.

[0005] For example, the patent document U.S. Pat. No. 11,400,200 “DEVICE FOR AUTOMATICALLY COLLECTING SNOT” discloses a device for automatically collecting snot, which comprises a suction nozzle, a collecting cup, a cup holder and a casing component which are detachably connected in sequence, wherein the casing component is a hollow cavity. The present disclosure generates negative pressure suction through the air pump, and then the suction air enters the inner collecting cup through the air suction pipe, the cup holder air suction port, the air suction port channel and the micropore in sequence. Finally, the suction sucks the snot in the nose of a user into the inner collecting cup through the suction nozzle. An air suction port channel is arranged opposite to the air suction port of the cup holder. The flow direction of the air channel changes. With the anti-reflux function of the cup cover, the problem that the nose is sucked back into the casing component and damages the host machine is effectively solved. Meanwhile, the cup cover is provided with micropores. This design can not only ensure the normal air intake effect, but also make use of the negative pressure of micropores to further prevent the nose from flowing backwards in the inner collecting cup. However, although the device for automatically collecting

nasal mucus solves the problem that the nasal mucus is sucked back into the casing component and damages the host computer, the device is not controlled by a constant pressure device, and the suction intensity of negative pressure generated by the air pump cannot be adjusted, so that the nasal cavity of infants may be injured when being used.
[0006] The nasal aspirator method disclosed in the patent document US 20150093268 “FOOT OPERATED SNIVEL SUCTION DEVICE” solves the problem that the negative pressure suction force cannot be adjusted. This document discloses a nasal aspirator, which comprises a base, two pumps, two pistons, two piston rods, a pedal and two casings. A pressure releasing valve is fixed in the casing with an air suction hole, and is connected between the air suction manifold and one of the air suction pipes. When the air suction pressure is too high, which may damage the nasal mucosa, the pressure releasing valve releases the pressure, and the sound generator sends out a warning sound, prompting the user to adjust the air suction pressure or check whether there is any wrong operation, so as to protect the user. Although this kind of pedal nasal aspirator solves the problem of air pump pressure, its structure is more complicated, and it still needs manual control for the change from the most traditional hand extrusion to pedal. Moreover, the air releasing valve deflates and discharges pressure after the pressure is higher than the normal pressure value, and it is not controlled at a stable pressure value, which still has certain safety risks during use.
SUMMARY
[0007] Accordingly, the present invention aims to provide a nasal aspirator that can automatically relieve pressure, so as to improve the above shortcomings.
[0008] The present invention provides a nasal aspirator, including:
[0009] a casing with an accommodating cavity, an air pump component, an air bag, a
[0010] constant pressure device and a suction component arranged in the accommodating cavity; and
[0011] wherein, the air pump component is communicated with the air bag and comprises at least one air flow connector, and the air flow connector defines an air suction channel and an air exhaust channel of the air pump component; and
[0012] wherein the air bag comprises at least one air chamber, and at least one communication member is arranged in the air chamber, and the communication member is configured to receive the air flow connector; and
[0013] wherein, the constant pressure device is arranged on the air bag and communicated with the air chamber, and comprises at least one pressure releasing valve; and the pressure releasing valve provides an overflow channel for air in the air chamber when the air in the air chamber exceeds a predetermined threshold; and
[0014] wherein the suction component is communicated with the air bag, and the suction component defines a path for external air to enter the air bag.
[0015] The present invention provides another nasal aspirator capable of automatically relieving pressure, which includes:
[0016] a casing with an accommodating cavity, an air pump component, an air bag, a constant pressure device and a suction component arranged in the accommodating cavity; and
[0017] wherein, the air pump component is communicated with the air bag and comprises at least one air flow connector, and the air flow connector defines an air suction channel and an air exhaust channel of the air pump component; and
[0018] wherein the air bag comprises at least one air chamber, and at least one communication member is arranged in the air chamber, and the communication member is configured to receive the air flow connector; and
[0019] wherein, the constant pressure device is arranged on the air bag and communicated with the air chamber, and comprises at least one pressure releasing valve; the pressure releasing valve provides an overflow channel for air in the air chamber when the air in the air chamber exceeds a predetermined threshold; and

[0020] wherein, the suction component defines an airflow path for external air to enter the air bag, and comprises a suction nozzle, a storage compartment and a hose, wherein the storage compartment is arranged between the suction nozzle and the hose, the storage compartment is configured to contain nasal mucus, and the hose is communicated with the air bag; and
[0021] wherein the storage compartment further comprises a pressure releasing channel and a switch component, and the switch component can open and close the pressure releasing channel.
[0022] The present invention also provides a method for sucking nasal mucus, which includes the following steps: providing a nasal aspirator capable of automatically relieving pressure, wherein the nasal aspirator comprises a casing with an accommodating cavity, an air pump component, an air bag, a constant pressure device and a suction component arranged in the accommodating cavity; and wherein, the air pump component is communicated with the air bag and comprises at least one air flow connector, and the air flow connector defines an air suction channel and an air exhaust channel of the air pump component; and wherein the air bag comprises at least one air chamber, at least one communication member is arranged in the air chamber, and the communication member is configured to receive the air flow connector; and wherein the constant pressure device is arranged on the air bag and communicated with the air chamber, and comprises at least one pressure releasing valve, and the pressure releasing valve provides an overflow channel for air in the air chamber when the air in the air chamber exceeds a predetermined threshold; and wherein the suction component is communicated with the air bag and comprises a suction nozzle and a storage compartment, and the suction component defines a path for external air to enter the air bag; and
[0023] opening the air pump component;
[0024] placing the suction nozzle close to a source of nasal mucus;
[0025] absorbing the nasal mucus and storing the nasal mucus in the storage compartment; and
[0026] closing the air pump component after absorbing the nasal mucus.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0027] In order to explain the technical solution of this application more clearly, the drawings needed in the implementation will be briefly introduced below. Obviously, the drawings described below are only some implementations of this application. For those skilled in the art, other drawings can be obtained according to these drawings without creative work.
[0028] FIG. **1** is an overall view of a nasal aspirator capable of automatically relieving pressure;
[0029] FIG. **2** is a cross-sectional view of the back of the casing of the nasal aspirator capable of automatically relieving pressure;
[0030] FIG. **3** is a partial view of the air pump component, air bag and constant pressure device in the nasal aspirator capable of automatically relieving pressure;
[0031] FIG. **4** is an exploded view of the front of the casing of the nasal aspirator capable of automatically relieving pressure;
[0032] FIG. **5** is a front view of the casing of the nasal aspirator capable of automatically relieving pressure;
[0033] FIG. **6** is a partial view of the nasal aspirator capable of automatically relieving pressure, including a hose and an air suction component;
[0034] FIG. **7** is a cross-sectional view of the lower compartment of the nasal aspirator capable of automatically relieving pressure;
[0035] FIG. **8** is a schematic view of the suction component in the nasal aspirator capable of automatically relieving pressure.
[0036] In the figures: Casing (**1000**); Music playing component (**1300**); Lighting component (**1400**); Electronic control module (**1500**); Control circuit board (**1510**); Control button (**1520**);

Power key (**1521**); Light key (**1522**); Music playing key (**1523**); Gear reduction key (**1524**); Gear increase key (**1525**); display panel (**1600**); Battery component (**1700**); Charging hole (**1800**); Air pump component (**2000**); Airflow connector (**2001**); Air suction connector (**2100**); Air exhaust connector (**2200**); Air bag (**3000**); Air chamber (**3001**); Air suction chamber (**3100**); Air suction port (**3110**); Suction communication member (**3120**); Air exhaust chamber (**3200**); A communication member: (**3002**); An air exhaust port (**3210**); An air exhaust Communication member (**3220**); Constant pressure device (**4000**); Pressure releasing valve (**4100**); Hose (**5000**); Suction component (**6000**); Storage compartment (**6100**); Upper compartment (**6110**); Suction nozzle mounting seat (**6111**); Lower compartment (**6120**); Slot (**6121**); Sealing ring (**6122**); Mounting seat (**6123**); Upper mounting seat (**6124**); Lower mounting seat (**6125**); Pressure releasing channel (**6201**); Switch component (**6200**); Suction nozzle (**6300**); Bracket (**6400**); First cover plate (**6410**); Second cover plate (**6420**); Through hole (**6430**); Airflow pipe (**6440**); Accommodating cavity (**7000**).

DESCRIPTION OF EMBODIMENTS

[0037] In describing the preferred embodiments, specific terminology will be resorted to for the sake of clarity. It is to be understood that each specific term includes all technical equivalents which operate in a similar manner to accomplish a similar purpose.

[0038] While various aspects and features of certain embodiments have been summarized above, the following detailed description illustrates a few exemplary embodiments in further detail to enable one skilled in the art to practice such embodiments. Reference will now be made in detail to embodiments of the inventive concept, examples of which are illustrated in the accompanying drawings. The accompanying drawings are not necessarily drawn to scale. The described examples are provided for illustrative purposes and are not intended to limit the scope of the invention. It should be understood, however, that persons having ordinary skill in the art may practice the inventive concept without these specific details.

[0039] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first attachment could be termed a second attachment, and, similarly, a second attachment could be termed a first attachment, without departing from the scope of the inventive concept.

[0040] It will be understood that when an element or layer is referred to as being “on,” “coupled to,” or “connected to” another element or layer, it can be directly on, directly coupled to or directly connected to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly coupled to,” or “directly connected to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0041] As used in the description of the inventive concept and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates other.

[0042] FIGS. **1** to **8** show a structure provided by a first embodiment of the present invention, and only parts related to the embodiment of the present invention are shown for convenience of explanation.

[0043] Referring to FIG. **1**, FIG. **1** discloses the general structure of the present invention. The nasal aspirator capable of automatically relieving pressure includes a casing **1000** with an accommodating cavity **7000**, and an air suction component **6000**, which includes a hose **5000** providing an air channel, a storage compartment **6100** and a suction nozzle **6300**. One end of the hose **5000** is detachably connected with the air suction port, and the other end is detachably connected with the lower mounting seat **6125**, and the hose **5000**, the air suction component **6000** and the air suction port are communicated.

[0044] Referring to FIG. **2** and FIG. **3**, an air pump component **2000** for providing negative pressure is arranged in the accommodating cavity **7000**, and the nasal aspirator capable of automatically releasing pressure can perform suction and air exhaust actions through the air pump component **2000**, and the air pump component **2000** is provided with an air flow connector **2001**, which includes an air suction connector **2100** and an air exhaust connector **2200**. In this embodiment, the accommodating cavity **7000** is also provided with an air bag **3000** connected with the air pump component **2000**, and the air bag **3000** is composed of an air suction chamber **3100** and an air exhaust chamber **3200** which are independent from each other. The air bag **3000** is provided with an air suction port **3110** and an air suction communication member **3120** communicating with the air suction chamber **3100**, an air exhaust port **3210** and an air exhaust communication member **3220** communicating with the air exhaust chamber **3200**. The suction connector **2100** of the air pump component **2000** is fixed on the air suction communication member **3120** and is in air communication with the air suction chamber **3100**, and the air exhaust connector **2200** of the air pump component **2000** is fixed on the air exhaust communication member **3220** and is in air communication with the air exhaust chamber **3200**.
[0045] When the nasal aspirator capable of automatically relieving pressure starts to work, the air pump component **2000** generates negative pressure, and is communicated with the outside through the air suction connector **2100**, the first connecting member, the air suction chamber **3100** and the air suction port **3110** in turn.
[0046] A constant pressure device **4000** for controlling the negative pressure is also arranged in the accommodating cavity **7000**, and air flows between the constant pressure device **4000** and the air suction chamber **3100**. A pressure releasing valve **4100** is arranged in the constant pressure device **4000**, and the pressure releasing valve **4100** is opened when the pressure in the inhaled chamber **3100** reaches a threshold; when the pressure in the suction chamber **3100** is lower than the threshold, the pressure releasing valve **4100** is closed.
[0047] A lighting component **1400** is provided at one side of the accommodation cavity **7000** in the longitudinal direction, which can provide lighting function. When it is needed to be used at night, the lighting in the room will feel dazzling, especially for infants. The lighting component **1400** can provide lights with different intensities, thus solving the above problems.
[0048] A music playing device is provided at the other side of the accommodating cavity **7000** in the length direction relative to the lighting component **1400**, and when in use, the music playing device can play, for example, gentle folk songs to attract the attention of infants, thus better completing the nose suction work.
[0049] There is also a battery module **1700** in the accommodating cavity **7000**. The battery module **1700** provides the working power of the nasal aspirator which can automatically relieve pressure, and the device is charged through a charging hole **1800** which is located under the opening **1100** of the casing **1000**. The charging hole **1800** can be a USB interface, a Type-c interface, a Lightning interface or other interfaces.
[0050] In another embodiment, the nasal aspirator is connected with a socket for power supply through an electrical connection device, such as an electrical plug or the like (not shown in the figure).
[0051] Referring to FIGS. **4** and **5**, FIG. **4** shows the front and internal structure of the casing **1000** of the nasal aspirator capable of automatically relieving pressure, and FIG. **5** shows the control button of the nasal aspirator capable of automatically relieving pressure.
[0052] As shown in FIG. **4**, an electronic control module **1500** is arranged in the accommodating cavity **7000**, which includes a control circuit board **1510** and a control button **1520**. The control circuit board **1510** is arranged in the accommodating cavity **7000**, and the control circuit board **1510** is electrically connected to the air pump component **2000**, the music playing component **1300** and the lighting component **1400**, and the control button **1520** is arranged on the control circuit board **1510**, penetrating the front of the casing **1000**.

[0053] The suction generated by the nasal aspirator capable of automatically relieving pressure is divided into **9** gears, which can be adjusted by the control button **1520**.
[0054] As shown in FIG. **5**, the control button **1520** includes: a power key **1521**, a light key **1522**, a music playing key **1523**, a gear reduction key **1524** and a gear increase key **1525**. In this embodiment, the power key **1521** is long-pressed to turn on and off the machine, and the default suction gear position is the **3**rd gear; the gear increase key **1525** is short-pressed to increase the gear, and the highest gear is the **9**th gear; the gear reduction key **1524** is short-pressed to decrease the gear, and the minimum gear is the **1**st gear; the power key **1521** is short-pressed to pause; the music playing key **1523** is short-pressed to cycle between low volume, medium volume, high volume and off. The light key is long-pressed to turn on or off the light, and is short-pressed to cycle between gear increase and gear reduction of the light intensity.
[0055] Before use, the nasal aspirator can be turned on, and the suction force of the suction nozzle **6300** can be felt by the palm before performing the nose suction work.
[0056] The front of the casing **1000** is also provided with a display panel **1600**, which is connected to the electronic control module **1500**. Through the display panel **1600**, the current working state of the nasal aspirator capable of automatically relieving pressure can be known, for example gear, power situation, whether the lights are on, light intensity, whether the music is on and volume.
[0057] As shown in FIGS. **6** and **7**, FIG. **6** shows the hose **5000** that can automatically relieve pressure and part of the suction component **6000**. FIG. **7** shows the cross section of the lower compartment **6120**. As shown in FIGS. **6** and **7**, the air suction component **6000** includes a storage compartment **6100** with a cavity **8000**, a suction nozzle **6300** fixed on the storage compartment **6100** and in air communication with the storage compartment **6100**, and a bracket **6400** detachably installed in the cavity **8000**.
[0058] The storage compartment **6100** is divided into an upper compartment **6110** and a lower compartment **6120**. The lower compartment **6120** is provided with a first mounting seat **6123**, which consists of an upper mounting seat **6124** and a lower mounting seat **6125**. The hose **5000** is connected with the lower compartment **6120** through the lower mounting seat **6125**, and the bracket **6400** is fixed with the lower compartment **6120** through the upper mounting seat **6124**.
[0059] The lower compartment **6120** is provided with a slot **6121**, and the slot **6121** is used for installing a sealing ring **6122**. When the upper compartment **6110** and the lower compartment **6120** are connected, the sealing ring **6122** can prevent nasal secretions sucked into the storage compartment **6100** from flowing out, and thus there is no need to worry about secretions overflowing or affecting the surrounding environment. The sealing ring **6122** is usually made of soft and durable materials to ensure its effectiveness and long-term use. It is precisely installed in the **6121** and closely adheres to the storage compartment **6100**, forming a reliable barrier to prevent any liquid or air from overflowing.
[0060] The bracket **6400** is provided with a first cover plate **6410**, which closes the end of the airflow pipe **6440** far from the upper mounting seat **6124**, and a second cover plate **6420** is provided below the first cover plate **6410**. The first cover plate **6410** and the second cover plate **6420** are both circular, and the diameter of the first cover plate **6410** is larger than that of the second cover plate **6420**. A through hole **6430** is opened between the first cover plate **6410** and the second cover plate **6420** for air communication between the storage compartment **6100** and the hose **5000**. The first cover plate **6410** is used to prevent the sucked nasal secretions from being sucked into the hose **5000**. Because the diameter of the first cover plate **6410** is larger than that of the second cover plate **6420**, when the nasal aspirator capable of automatically relieving pressure works, the sucked nasal secretions will not fall on the second cover plate **6420** when they fall on the first cover plate **6410** and disperse around, but fall on the lower end of the storage compartment **6100**, and the second cover plate **6420** will also prevent the nasal secretions sucked into the storage compartment **6100** from entering the through hole **6430** from below by suction.
[0061] At least a part of the storage compartment **6100** is made of a transparent material, and users

can directly observe the internal situation of the storage compartment **6100** outside the storage compartment **6100** without opening or moving articles, and the storage compartment constructed of a transparent material provides visual convenience.

[0062] Suction nozzle **6300** can be made of medical grade silica gel, which ensures the safety and reliability of the product, does not contain any harmful substances, is non-toxic and tasteless, and meets the international hygiene standards through strict production technology and testing standards. The suction nozzle **6300** has softness and elasticity. Because the silica gel material itself is not easy to adhere to stains or bacteria, it can be cleaned with clean water after daily use. Moreover, the suction nozzle **6300** can remain intact after repeated tensile tests and tear resistance tests, and it also gives infants a comfortable experience.

[0063] The suction nozzle **6300** is fixed to the upper compartment **6110** through the suction nozzle mounting seat **6111** provided in the upper compartment **6110**. When in use, the suction nozzle **6300** is placed close to the hand palm, and the appropriate suction gear is adjusted by the control button **1520** to perform a nasal aspirator, and the suction nozzle **6300** is placed at the nostril without extending into the nasal cavity. Before using the nasal aspirator capable of automatically relieving pressure, a normal saline can be dripped to soften nasal secretions. After suction, the storage compartment **6100** can be opened and cleaned with clean water.

[0064] As shown in FIG. **8**, FIG. **8** shows another embodiment of the nasal aspirator capable of automatically relieving pressure. The storage compartment **6100** in the suction component **6000** is provided with a pressure releasing channel **6201** and a switch component **6200**. The switch component **6200** controls the opening and closing of the pressure releasing channel **6201**. When the pressure releasing channel **6201** is closed, a negative pressure space is formed, and the air pump component **2000** works to generate negative pressure to form suction, and after passing through the air bag **3000**, the hose **5000**, the storage compartment **6100** and the suction nozzle **6300** in turn, suction is generated at the suction nozzle **6300** and communicated with the outside. When the pressure releasing channel **6201** is opened, an additional pressure releasing channel is provided on the storage compartment **6100**, which makes the whole nasal aspirator safer.

[0065] In another embodiment, the nasal aspirator has a casing **1000** containing a cavity **7000**, an air pump component **2000**, an air bag **3000**, a constant pressure device **4000** and a suction component **6000** arranged in the cavity **7000**. The suction component **6000** includes a storage compartment **6100**, and a suction nozzle **6300**, that is, the suction component **6000** does not include a hose **5000** (not shown in the figure).

[0066] In another embodiment, the nasal aspirator includes a casing **1000** with an accommodating cavity **7000**, an air pump component **2000**, an air bag **3000**, a constant pressure device **4000** and a suction component **6000** arranged in the accommodating cavity **7000**, wherein the air pump component **2000** includes an air flow connector **2001**, and the air bag **3000** includes an air chamber **3001**, and the constant pressure device **4000** is arranged on the air bag **3000** and communicates with the air chamber **3001** (not shown in the figure).

[0067] In another embodiment, the number of constant pressure devices **4000** is two. In other embodiments, the number of constant pressure devices **4000** can be any desired number.

[0068] In another embodiment, the number of air bags **3000** is two, and in other embodiments, the number of air bags **3000** can be any desired number.

[0069] In another embodiment, the air suction component **6000** can be used once and does not need to be cleaned after use, thus avoiding the possibility of virus infection due to improper cleaning. Because it is impossible to control whether the user performs the cleaning operation correctly or not, it may become a source of bacteria over a long period of time. However, by providing disposable substitutes, the spread of bacteria can be effectively prevented and the health and safety of users can be guaranteed.

[0070] In another embodiment, the nasal aspirator does not include a lighting component **1400** or a music component **1300** (not shown in the figure).

[0071] The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. The use of “adapted to” or “configured to” herein is meant as open and inclusive language that does not foreclose devices adapted to or configured to perform additional tasks or steps. Additionally, the use of “based on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based on” one or more recited conditions or values may, in practice, be based on additional conditions or values beyond those recited. Similarly, the use of “based at least in part on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based at least in part on” one or more recited conditions or values may, in practice, be based on additional conditions or values beyond those recited. Headings, lists, and numbering included herein are for ease of explanation only and are not meant to be limiting.
[0072] The various features and processes described above may be used independently of one another, or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of the present disclosure. In addition, certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be performed in other sequences that are appropriate. For example, described blocks or states may be performed in an order other than that specifically disclosed, or multiple blocks or states may be combined in a single block or state. The example blocks or states may be performed in serial, in parallel, or in some other manner. Blocks or states may be added to or removed from the disclosed examples. Similarly, the example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed examples.
[0073] The invention has now been described in detail for the purposes of clarity and understanding. However, those skilled in the art will appreciate that certain changes and modifications may be practiced within the scope of the appended claims. [0074] Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain examples include, while other examples do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more examples or that one or more examples necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular example.

Claims

1. A nasal aspirator capable of automatically relieving pressure, comprising: a casing with an accommodating cavity, an air pump component, an air bag, a constant pressure device and a suction component arranged in said accommodating cavity; and wherein, said air pump component is communicated with said air bag and comprises at least one air flow connector, and said air flow connector defines an air suction channel and an air exhaust channel of said air pump component; and wherein said air bag comprises at least one air chamber, and at least one communication member is arranged in said air chamber, and said communication member is configured to receive said air flow connector; and wherein, said constant pressure device is arranged on said air bag and communicated with said air chamber, and comprises at least one pressure releasing valve; and the pressure releasing valve provides an overflow channel for air in said air chamber when the air in said air chamber exceeds a predetermined threshold; and wherein said suction component is

communicated with said air bag, and said suction component defines a path for external air to enter said air bag.
2. The nasal aspirator according to claim 1, wherein said air flow connector comprises an air suction connector and an air exhaust connector; and said air chamber comprises an air suction chamber and an air exhaust chamber, and said communication member comprises an air suction communication member and an air exhaust communication member; and said air suction connector is arranged on said air suction communication member and communicates with said air suction chamber, and said air exhaust connector is arranged on said air exhaust communication member and communicates with said air exhaust chamber.
3. The nasal aspirator according to claim 2, wherein said constant pressure device is communicated with said air suction chamber, and when said pressure in said air suction chamber reaches a threshold, said pressure releasing valve is opened; and when said pressure in said suction chamber is lower than the threshold, said pressure releasing valve is closed.
4. The nasal aspirator according to claim 3, wherein said casing is provided with an air suction port and an air exhaust port communicated with said air bag, wherein said air suction port is communicated with said air suction chamber, and said air exhaust port is communicated with said air exhaust chamber.
5. The nasal aspirator according to claim 1, wherein said air suction component comprises a hose, a storage compartment and a suction nozzle, wherein one end of said storage compartment is communicated with said hose and the other end is communicated with said suction nozzle, and a channel for air to flow is formed between said storage compartment and said hose.
6. The nasal aspirator according to claim 5, wherein said storage compartment comprises an upper compartment, a lower compartment, a sealing ring and a bracket; and said lower compartment is also provided with a slot, and said sealing ring is fixed on said slot.
7. The nasal aspirator according to claim 6, wherein said lower compartment is provided with a mounting seat, which comprises an upper mounting seat and a lower mounting seat, said bracket is fixed on said upper mounting seat, and said hose is fixed on said lower mounting seat.
8. The nasal aspirator according to claim 6, wherein said upper compartment is provided with a suction nozzle mounting seat, and said suction nozzle is fixed on said suction nozzle mounting seat.
9. The nasal aspirator according to claim 7, wherein said bracket comprises a first cover plate, a second cover plate arranged below said first cover plate and an air flow pipe; and wherein that first cover plate and said second cover plate are circular, a diameter of said first cover plate is larger than that of said second cover plate, and a through hole is also arranged on said air flow pipe, and said through hole is positioned between said first cover plate and said second cover plate.
10. The nasal aspirator according to claim 6, wherein at least a part of said storage compartment is made of a transparent material.
11. The nasal aspirator according to claim 1, wherein a battery component, a music playing component and a lighting component are also arranged in said accommodating cavity.
12. The nasal aspirator according to claim 11, wherein an electronic control module is further arranged in said accommodating cavity, and said electronic control module comprises a control circuit board and a control button, wherein said control circuit board is electrically connected to said air pump component, said music playing component and said lighting component, and said control button is arranged on said control circuit board.
13. The nasal aspirator according to claim 12, wherein said nasal aspirator has different suction gears and is controlled by said control button.
14. The nasal aspirator according to claim 12, wherein said casing is provided with a display panel, which is connected to said electronic control module and used for displaying a current working state of said nasal aspirator.
15. A nasal aspirator capable of automatically relieving pressure, comprising: a casing with an accommodating cavity, an air pump component, an air bag, a constant pressure device and a suction

component arranged in said accommodating cavity; and wherein, said air pump component is communicated with said air bag and comprises at least one air flow connector, and said air flow connector defines an air suction channel and an air exhaust channel of said air pump component; and wherein said air bag comprises at least one air chamber, and at least one communication member is arranged in said air chamber, and said communication member is configured to receive said air flow connector; and wherein, said constant pressure device is arranged on said air bag and communicated with said air chamber, and comprises at least one pressure releasing valve; and the pressure releasing valve provides an overflow channel for air in said air chamber when the air in said air chamber exceeds a predetermined threshold; and wherein, said suction component defines an airflow path for external air to enter said air bag, and comprises a suction nozzle, a storage compartment and a hose, wherein said storage compartment is arranged between said suction nozzle and said hose, said storage compartment is configured to contain nasal mucus, and said hose is communicated with said air bag; and wherein said storage compartment further comprises a pressure releasing channel and a switch component, and said switch component can open and close said pressure releasing channel.

16. A method for sucking nasal mucus, comprising the following steps: providing a nasal aspirator capable of automatically relieving pressure, wherein said nasal aspirator comprises a casing with an accommodating cavity, an air pump component, an air bag, a constant pressure device and a suction component arranged in said accommodating cavity; and wherein, said air pump component is communicated with said air bag and comprises at least one air flow connector, and said air flow connector defines an air suction channel and an air exhaust channel of said air pump component; and wherein said air bag comprises at least one air chamber, at least one communication member is arranged in said air chamber, and said communication member is configured to receive said air flow connector; and wherein said constant pressure device is arranged on said air bag and communicated with said air chamber, and comprises at least one pressure releasing valve, and said pressure releasing valve provides an overflow channel for air in said air chamber when the air in said air chamber exceeds a predetermined threshold; and wherein said suction component is communicated with said air bag and comprises a suction nozzle and a storage compartment, and said suction component defines a path for external air to enter said air bag; and opening said air pump component; placing said suction nozzle close to a source of nasal mucus; absorbing said nasal mucus and storing said nasal mucus in said storage compartment; and closing said air pump component after absorbing said nasal mucus.

17. The method according to claim 16, wherein before placing said suction nozzle near the source of nasal mucus, said suction nozzle is put near a hand palm to adjust an appropriate suction gear before sucking the nasal mucus.

18. The method according to claim 16, wherein when said suction nozzle is placed close to said source of nasal mucus, said suction nozzle is placed at a nostril without going deep into a nasal cavity.

19. The method according to claim 16, wherein before placing said suction nozzle near the source of nasal mucus, a normal saline is used to soften the nasal mucus.

20. The method according to claim 16, wherein after closing said air pump component, said storage compartment is opened and cleaned.